a1) $w(x,t) = \sin(x+ct)$

für t

$$\frac{\partial w}{\partial t} = \frac{\partial}{\partial t}\sin(x+ct) = c\cos(x+ct)$$

$$\frac{\partial^2 w}{\partial t^2} = \frac{\partial}{\partial t}\left(c\cos(x+ct)\right) = -c^2\sin(x+ct)$$

für x

$$\frac{\partial w}{\partial x} = \frac{\partial}{\partial x}\sin(x+ct) = \cos(x+ct)$$

$$\frac{\partial^2 w}{\partial x^2} = \frac{\partial}{\partial x}\left(\cos(x+ct)\right) = -\sin(x+ct)$$

Wellengleichung:

$$\frac{\partial^2 w}{\partial t^2} = c^2\frac{\partial^2 w}{\partial x^2} \quad \Rightarrow \quad -c^2\sin(x+ct) = c^2\cdot\left(-\sin(x+ct)\right)$$

a2) $v(x,t) = \sin(x+ct) + \cos(2x+2ct)$

für t

$$\frac{\partial w}{\partial t} = \frac{\partial}{\partial t}\sin(x+ct) + \frac{\partial}{\partial t}\cos(2x+2ct) = c\cos(x+ct) + \left(-2c\sin(2x+2ct)\right)$$

$$\frac{\partial^2 w}{\partial t^2} = \frac{\partial}{\partial t}c\cos(x+ct) + \frac{\partial}{\partial t}\left(-2c\sin(2x+2ct)\right) = -c^2\sin(x+ct) - 4c^2\cos(2x+ct)$$

für x

$$\frac{\partial w}{\partial x} = \frac{\partial}{\partial x}\sin(x+ct) + \frac{\partial}{\partial x}\cos(2x+2ct) = \cos(x+ct) + \left(-2\sin(2x+2ct)\right)$$

$$\frac{\partial^2 w}{\partial x^2} = \frac{\partial}{\partial x}\cos(x+ct) + \frac{\partial}{\partial x}\left(-2\sin(2x+2ct)\right) = -\sin(x+ct) - 4\cos(2x+2ct)$$

Wellengleichung:

$$\frac{\partial^2 w}{\partial t^2} = c^2\frac{\partial^2 w}{\partial x^2} \quad \Rightarrow$$

$$-c^2\sin(x+ct) - 4c^2\cos(2x+ct) = c^2\left(-\sin(x+ct) - 4\cos(2x+2ct)\right)$$
$$= -c^2\sin(x+ct) - 4c^2\cos(2x+ct)$$